Task 11: Indexing and Query Performance Optimization

Tools:

- MySQL Workbench
- Alternatives: PostgreSQL, BigQuery Sandbox, SQLite

Hints / Mini Guide:

1. Identify slow-running queries using large datasets.
2. Execute queries without indexes and note execution time.
3. Create indexes on frequently searched columns.
4. Re-run queries and compare performance.
5. Use EXPLAIN to analyze query execution plans.
6. Understand clustered vs non-clustered indexes.
7. Identify cases where indexes hurt performance.
8. Document best indexing practices.

Deliverables:

- SQL file with index creation and EXPLAIN outputs

Final Outcome:

- Intern understands SQL performance tuning fundamentals.

Interview Questions Related To Above Task:

- What is an index?
- Why indexes improve performance?
- When should indexes be avoided?
- What is EXPLAIN used for?
- Difference between clustered and non-clustered index?

📌 **Task Submission Guidelines**

- ⏰ **Time Window:**

Youcan complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

Youare free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🛠️ **Debug Yourself:**

Tryto resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💸 **No Paid Tools:**

Ifthe task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.
Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

📤 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [Submission Link]